

Cybersecurity Incident Report: Network Traffic Investigation

Analyst Information

Analyst: Nitin Sukthe

Role: SOC Analyst (Lab Simulation)

Investigation Type: Network Traffic Analysis

Tool Used: Wireshark

Environment: Kali Linux Virtual Machine

Capture File: Packet_capture.pcapng

Part 1: Summary of the Observed Network Activity

As part of routine network monitoring activities, live network traffic was captured and analyzed using Wireshark. The purpose of the investigation was to validate network communications generated during web browsing activity and determine whether any indicators of compromise or suspicious traffic patterns were present.

During the packet capture, browser traffic was generated by accessing:

- <https://www.kali.org>
- <https://www.redhat.com/en>

Analysis of the packet capture identified the following protocols:

- DNS
- TCP
- TLS/SSL
- HTTPS
- IPv4

DNS traffic was observed prior to encrypted communications. TCP sessions were established successfully, followed by TLS handshakes and HTTPS communications between the local workstation and external web servers.

No packet-level evidence of malicious communications, unauthorized remote connections, command-and-control traffic, or data exfiltration was identified during the investigation.

Evidence Identified During Analysis

DNS Activity

A DNS display filter was applied to isolate domain resolution traffic.

Observed behavior included:

- DNS query requests
- DNS response packets
- Domain-to-IP address resolution

DNS communications occurred before HTTPS sessions were established, which is expected behavior for web browsing operations.

Protocol Hierarchy Findings

Protocol Hierarchy Statistics showed that the majority of traffic consisted of:

- TCP communications
- TLS encrypted traffic
- HTTPS web traffic
- DNS requests

The protocol distribution aligned with normal browser-generated communications.

Conversation Statistics Findings

Conversation analysis identified active communications between the analyst workstation and external web infrastructure.

Traffic volume analysis revealed that the highest-byte conversations were associated with browser sessions generated during the investigation.

No unauthorized peer-to-peer communications or unusual destinations were observed.

Part 2: Investigation Analysis and Incident Assessment

Initial Observation

The network investigation was initiated after collecting live traffic generated through web browsing activity.

The objective was to determine whether captured communications represented legitimate user activity or potential security concerns.

Packet capture analysis confirmed:

- Successful DNS resolution
- Normal TCP session establishment
- Valid TLS handshakes
- Encrypted HTTPS communications

These findings indicate expected client-server communication behavior.

Current Investigation Status

The investigation is currently classified as:

Status: Closed – No Malicious Activity Identified

Evidence collected during packet analysis does not indicate:

- Malware communications
- Command-and-Control activity
- DNS tunneling
- Data exfiltration
- Unauthorized outbound connections
- Network scanning activity

Based on available evidence, the observed traffic is considered legitimate browser-generated network activity.

Security Assessment

Although no active security incident was identified, the investigation highlighted several areas that analysts should continuously monitor.

Potential indicators requiring additional investigation include:

DNS Indicators

- Excessive DNS requests
- Newly registered domains
- Suspicious domain patterns
- DNS tunneling behavior

Network Indicators

- Unknown external IP addresses
- Unusual destination countries
- Persistent encrypted sessions
- Abnormal outbound traffic volumes

Endpoint Indicators

- Suspicious processes
 - Unauthorized scheduled tasks
 - Unexpected network connections
 - Persistence mechanisms
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Recommended Next Steps

The following actions are recommended as part of ongoing network monitoring activities:

Continue Traffic Monitoring

Monitor:

- DNS activity
- HTTPS communications
- External connections

for unusual behavioral changes.

Validate Destination Reputation

Verify external destinations using:

- Threat intelligence feeds
- Domain reputation services
- IP reputation databases

Correlate Security Events

Compare packet capture findings against:

- Firewall logs
- Endpoint logs
- SIEM alerts
- Authentication records

to identify related activity.

Maintain Evidence

Preserve:

- Packet captures
- Screenshots
- Investigation notes
- Security logs

to support future investigations and training activities.

Suspected Root Cause

No security incident was identified during this investigation.

Observed network traffic was generated by legitimate browser interactions with trusted external websites.

The captured communications were consistent with normal web browsing behavior and did not require escalation to the incident response team.

SOC Analyst Conclusion

The network traffic investigation successfully validated browser-generated communications captured through Wireshark.

Analysis of DNS traffic, protocol hierarchy statistics, and conversation statistics confirmed that the observed activity represented normal encrypted web communications.

No indicators of compromise were identified during the assessment.

This investigation demonstrates practical SOC analyst capabilities including:

- Packet Capture Analysis
- DNS Investigation
- Protocol Analysis
- Network Monitoring
- Security Event Validation
- Incident Assessment
- Security Documentation

Final Assessment:

No active security incident detected. Traffic behavior consistent with legitimate web browsing activity.